# Engineering Mechanics
# Lecture- II

By
Dr. Aali Pant
Assistant Professor
Civil and Infrastructure Engineering
Indian Institute of Technology Jodhpur

# Force Systems

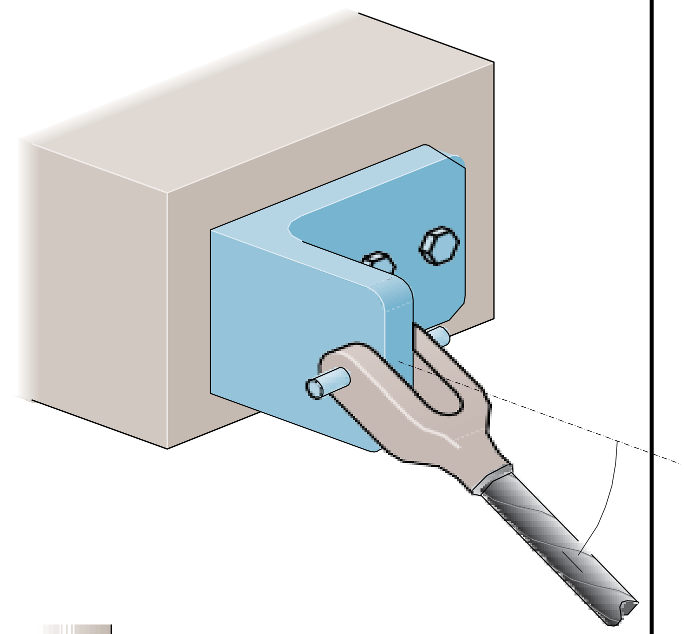
## Force

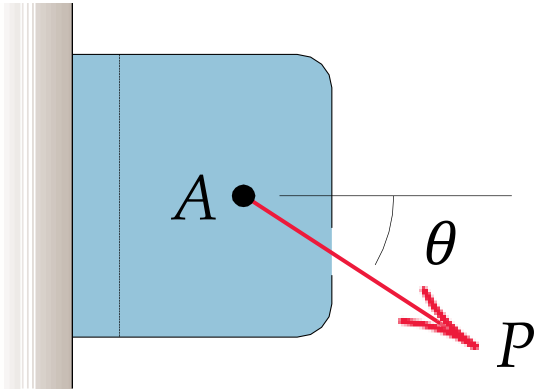
- Force: an action of one body on another

- force is a vector quantity, because its effect depends on the **direction** as well as on the **magnitude** of the action

- the complete specification of the action of a force must include its ***magnitude***, ***direction***, and ***point of application.***

- External and Internal Effects



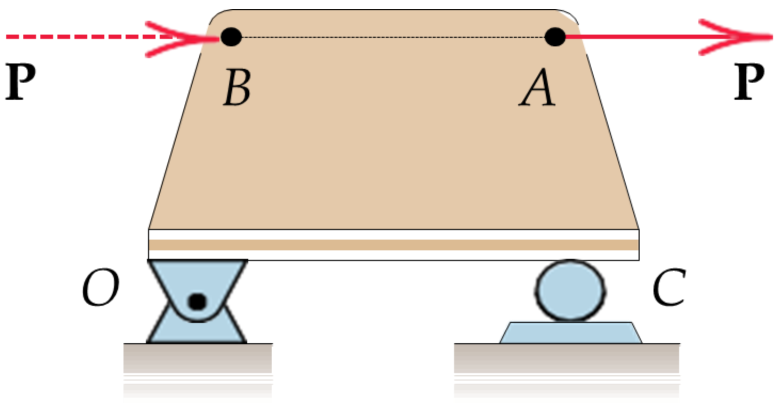


## Principle of Transmissibility

- A force may be applied at any point on its given line of action without altering the resultant effects of the force external to the rigid body on which it acts.
- Force may be considered as Sliding vector

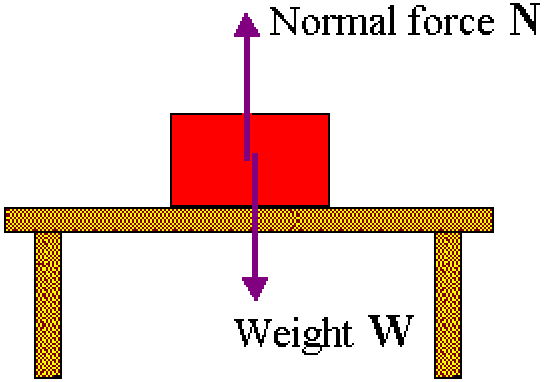
- The force **P** acting on the rigid plate in may be applied at *A* or at *B* or at any other point on its line of action, and the net external effects of **P** on the bracket will not change.

- The external effects are the force exerted on the plate by the bearing support at *O* and the force exerted on the plate by the roller support at *C*.



# Force Classification

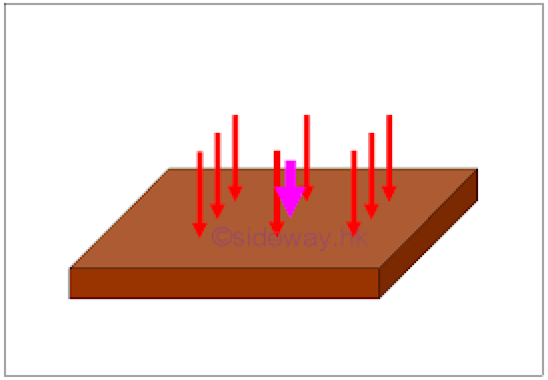
- **Contact Force** is produced by direct physical contact, e.g., force exerted on a body by a supporting surface



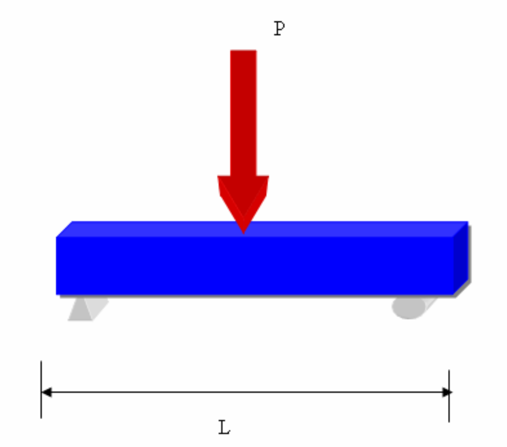

- **Body Force**: is generated by virtue of the position of a body within a force fieldsuch as a gravitational, electric, or magnetic field. e.g., weight

# Force Classification

- **Distributed force**: Force distributed over a region.
  - Every contact force is actually applied over a finite area and is therefore really a distributed force.




- **Concentrated force**: Force concentrated at a point
  - when the dimensions of the area are very small compared with the other dimensions of the body, we may consider the force to be concentrated at a point with negligible loss of accuracy.

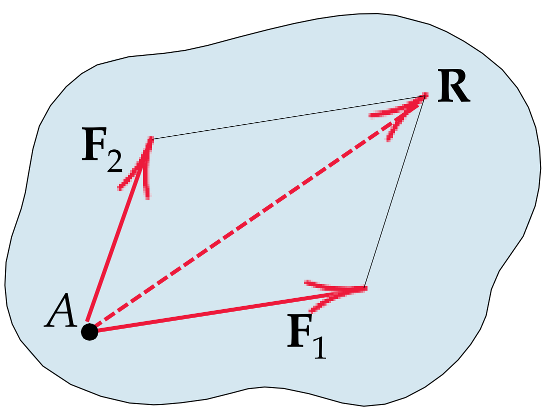


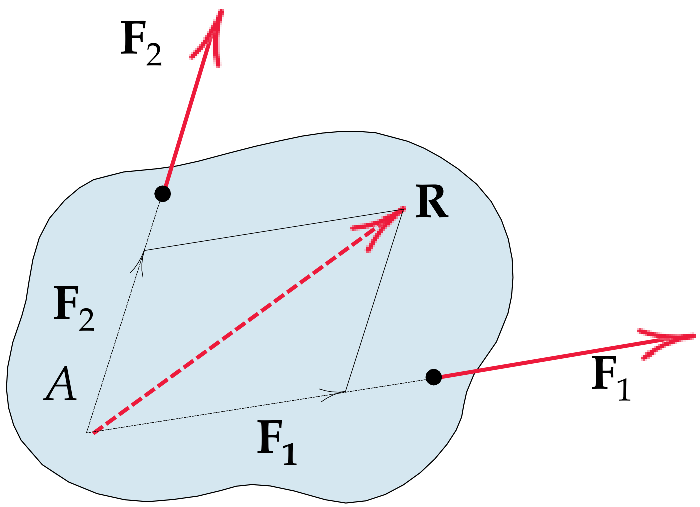
# Action and Reaction

- According to Newton's third law, the action of a force is always accompanied by an equal and opposite reaction.
- Distinguish between the action and the reaction in a pair of forces.
- we first isolate the body in question and then identify the force exerted on that body (not the force exerted by the body).
- It is very easy to mistakenly use the wrong force of the pair unless we distinguish carefully between action and reaction.

# Concurrent Forces

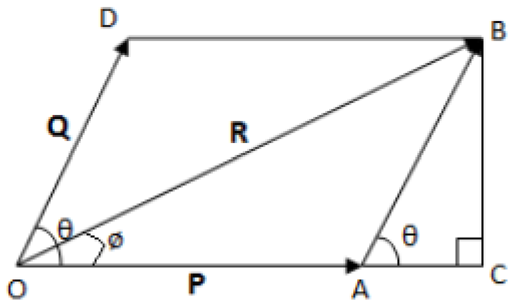
- Two or more forces are said to be concurrent at a point if their lines of action intersect at that point.



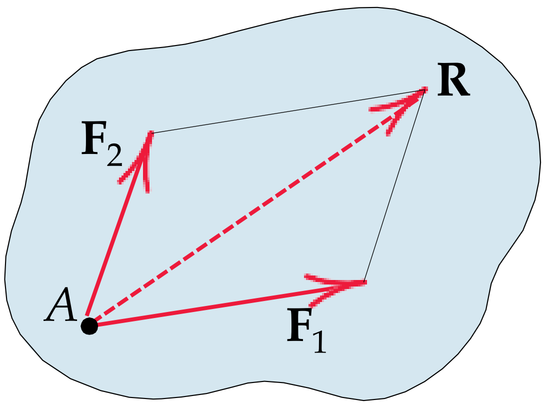



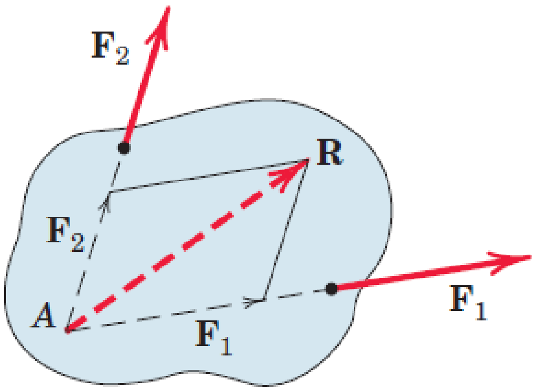
- **PARALLELOGRAM LAW OF FORCES**

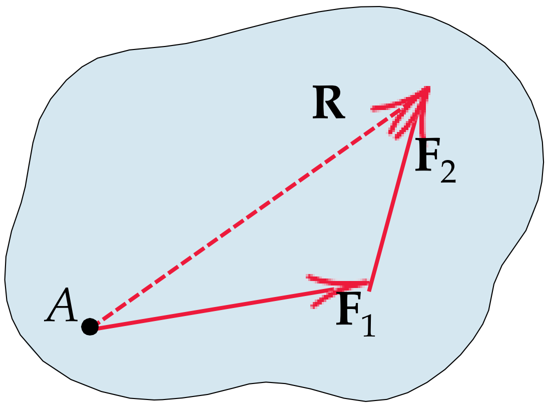
- It states that if two concurrent forces, acting simultaneously on a body be represented in magnitude and direction by the two sides of a parallelogram then their resultant may be represented in magnitude and direction by the diagonal of the parallelogram, drawn from the same point.



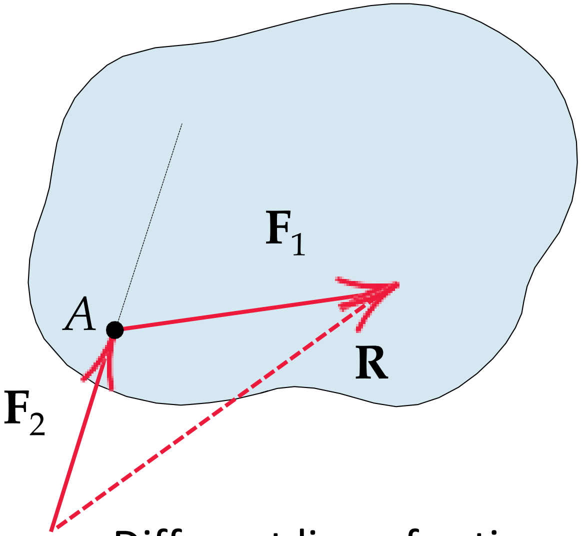
# Concurrent Forces




Same line of action

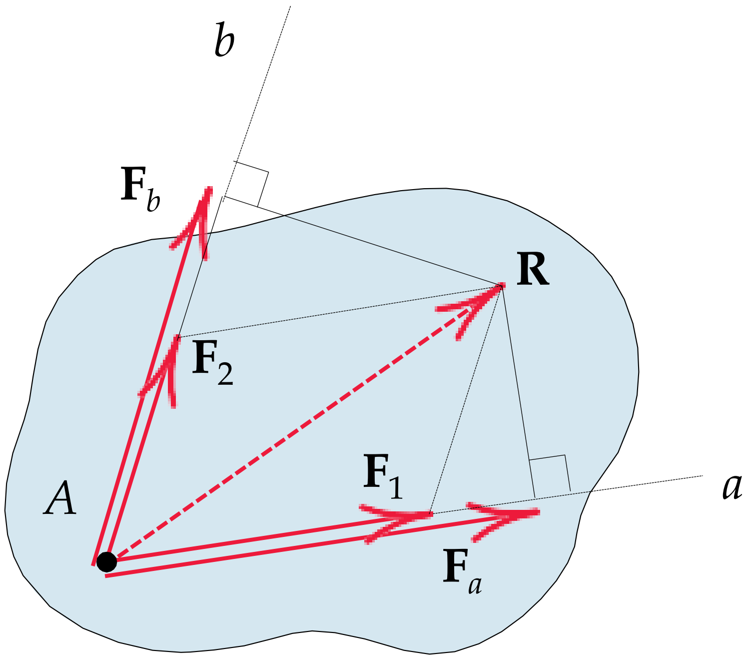



Different line of action

## Vector Components

- we often need to replace a force by its vector components in directions which are convenient for a given application.

$$\boxed{\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2}$$




**Components vs Projections**

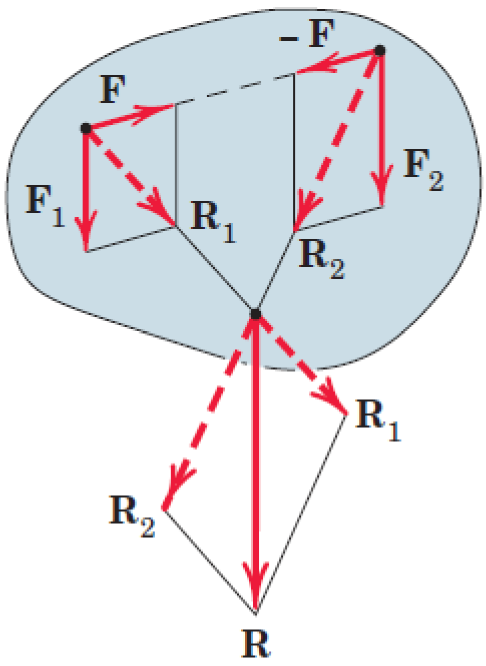
- **A Special Case of Vector Addition**

- The two vectors are combined by first adding two equal, opposite, and collinear forces.

- Adding **F**1 and **F** to produce **R**1

- Adding F2 and –F to Produce R2

- Solving R1 and R2 yields R



# RECTANGULAR COMPONENTS

- The most common two-dimensional resolution of a force vector is into rectangular components.

$$\boxed{\mathbf{F} = \mathbf{F}_x + \mathbf{F}_y}$$

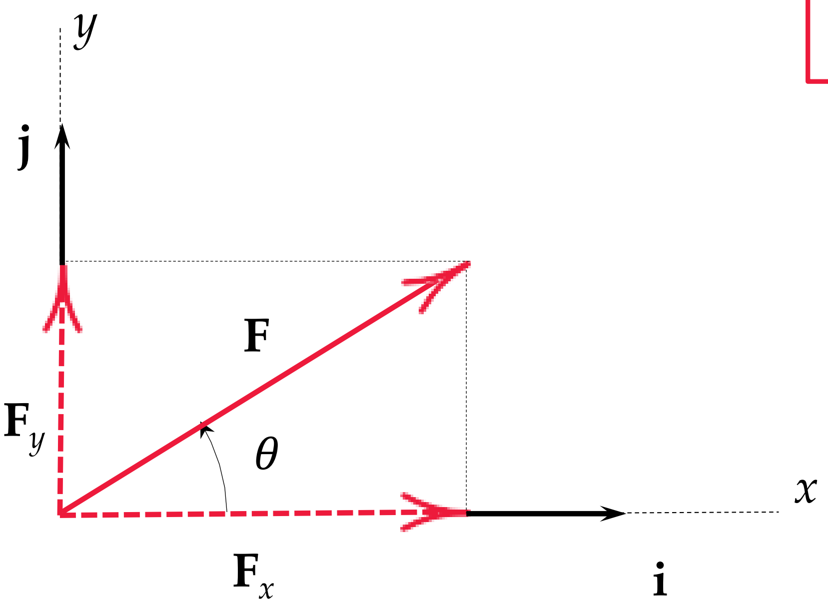
$$\boxed{\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j}}$$

$$\boxed{F = \sqrt{(F_x)^2 + (F_x)^2}}$$

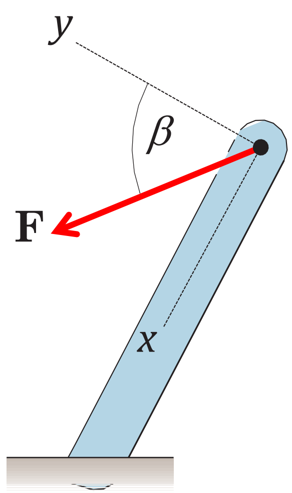
$$\boxed{F_x = F\cos\theta}$$

$$\boxed{F_y = F\sin\theta}$$

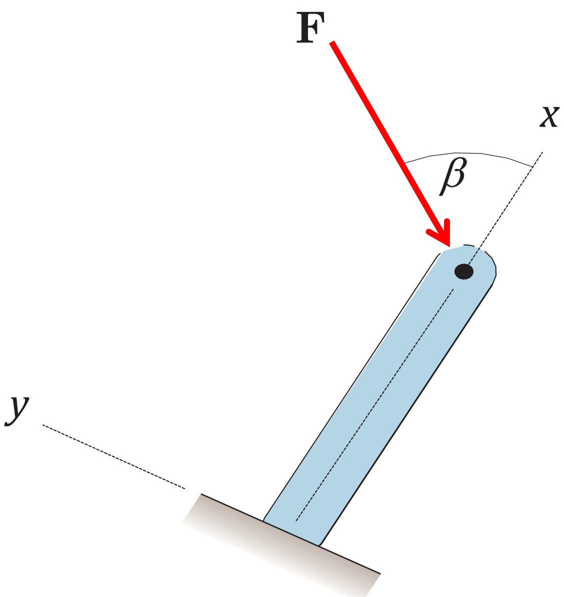
$$\boxed{\theta = \tan^{-1}(F_y/F_x)}$$



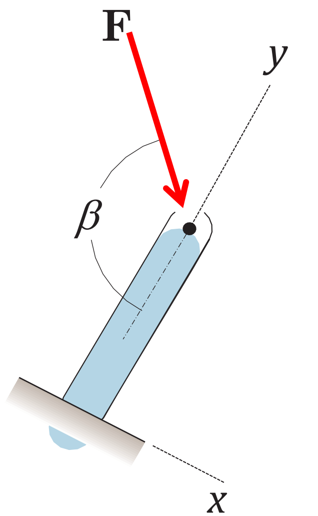
# Determining the Components of a Force



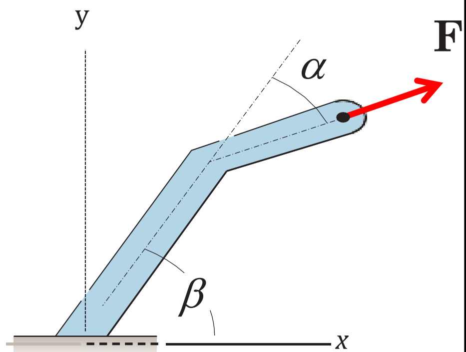

$$F_x = F \sin\beta$$
$$F_y = F \cos\beta$$




$$F_x = -F\cos\beta$$
$$F_y = -F\sin\beta$$

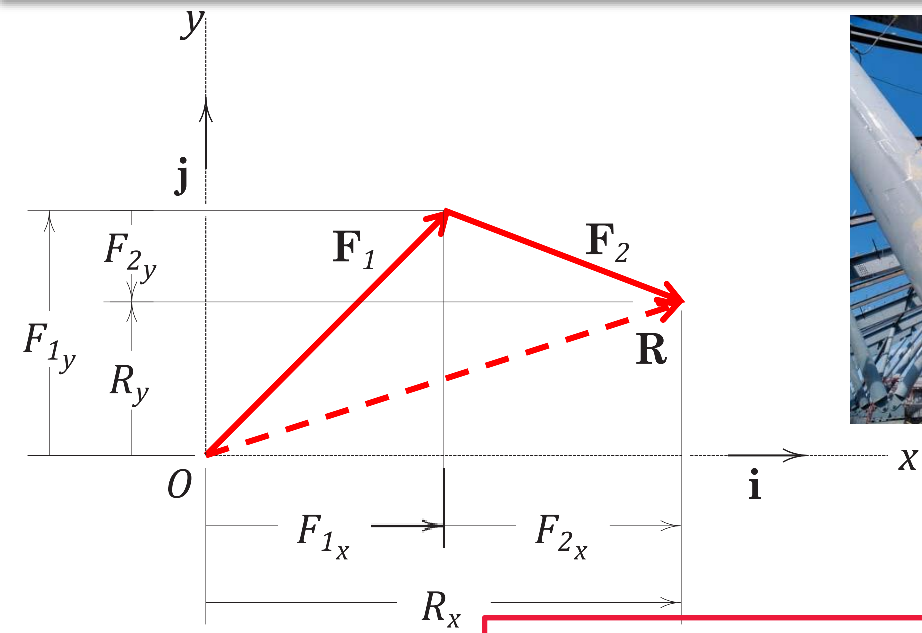



$$F_x = F\sin(\pi - \beta)$$
$$F_y = -F\cos(\pi - \beta)$$




$$F_x = F\cos(\beta - \alpha)$$
$$F_y = F\sin(\beta - \alpha)$$

# Determining the Components of Force




$$\mathbf{F}_1 = F_{1x}\mathbf{i} + F_{1y}\mathbf{j}$$

$$\mathbf{F}_2 = F_{2x}\mathbf{i} + F_{2y}\mathbf{j}$$

$$\mathbf{R} = R_x\mathbf{i} + R_y\mathbf{j}$$

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_1$$

$$R_x\mathbf{i} + R_y\mathbf{j} = (F_{1x}\mathbf{i} + F_{1y}\mathbf{j}) + (F_{2x}\mathbf{i} + F_{2y}\mathbf{j})$$

$$R_x\mathbf{i} + R_y\mathbf{j} = (F_{1x} + F_{2x})\mathbf{i} + (F_{1y} + F_{2y})\mathbf{j}$$

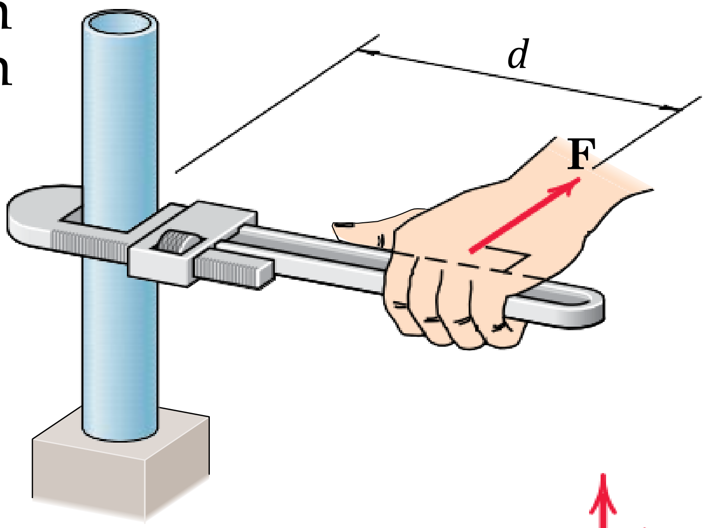
$$R_x = (F_{1x} + F_{2x}) = \sum F_{ix}$$

$$R_y = \left(F_{1y} + F_{2y}\right) = \sum F_{iy}$$

# Moment

- Force can tend to rotate a body about an axis. This rotational tendency is known as the moment M of the force.

- Moment is also referred to as torque.

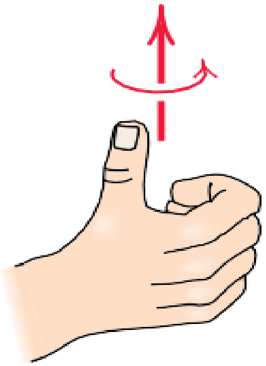



(a)

**Magnitude of moment**

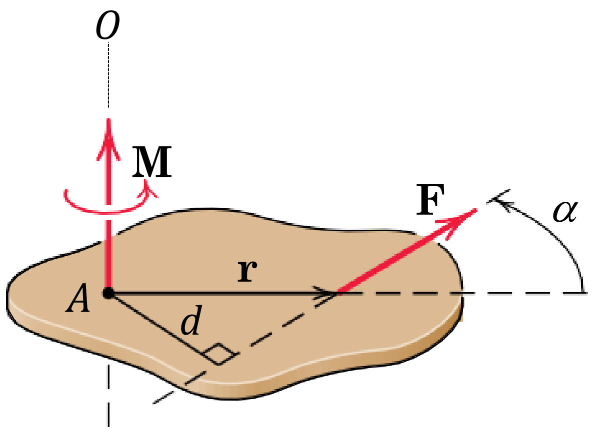
$$\boxed{M = Fd}$$

**Direction**

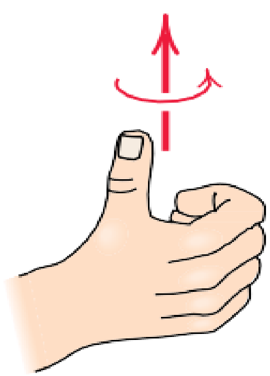


(c)

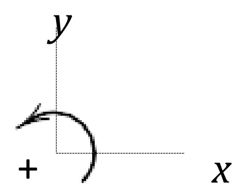
# General 2D case



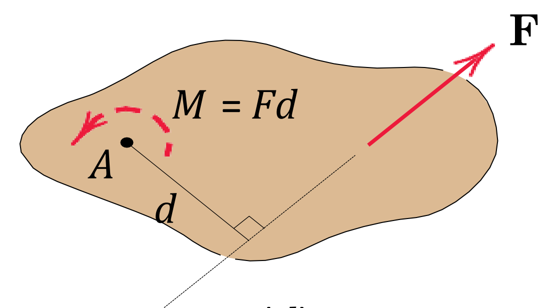

(b)



(c)




Anti clockwise +ve




(d)

$$\boxed{M = Fd}$$

$$\boxed{d = r\sin\alpha}$$

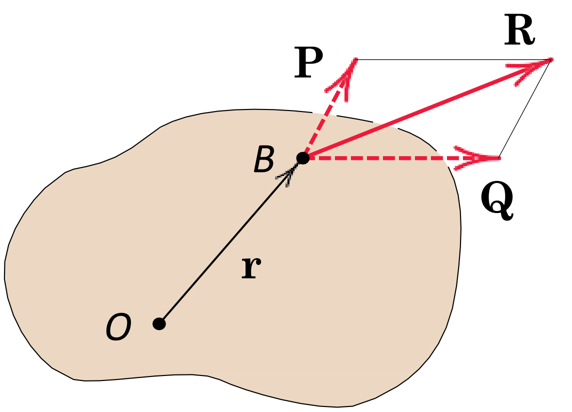
$$\boxed{M = Fr\sin\alpha}$$

$$\boxed{M = |\mathbf{r} \times \mathbf{F}|}$$

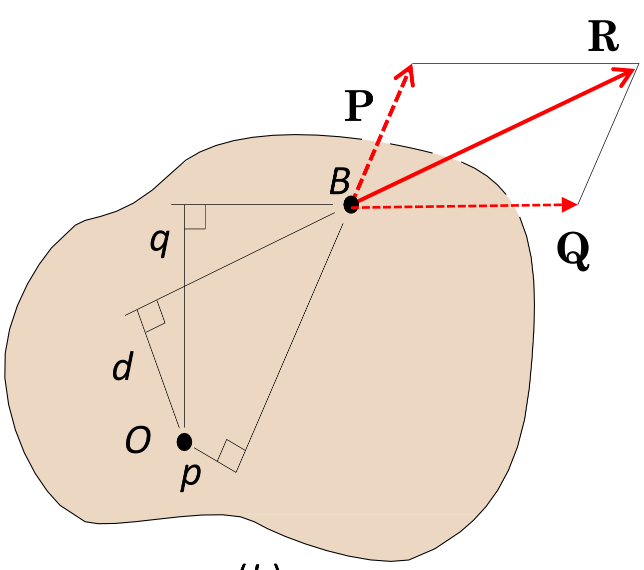
$$\boxed{\mathbf{M} = \mathbf{r} \times \mathbf{F}}$$

# Varignon's Theorem

- The moment of a force about any point is equal to the sum of the moments of the components of the force about the same point.




(*a*)




(*b*)

# Varignon's Theorem

$$\boxed{\mathbf{M}_O = \mathbf{r} \times \mathbf{R}}$$

$$\boxed{\mathbf{M}_O = \mathbf{r} \times (\mathbf{P} + \mathbf{Q})}$$

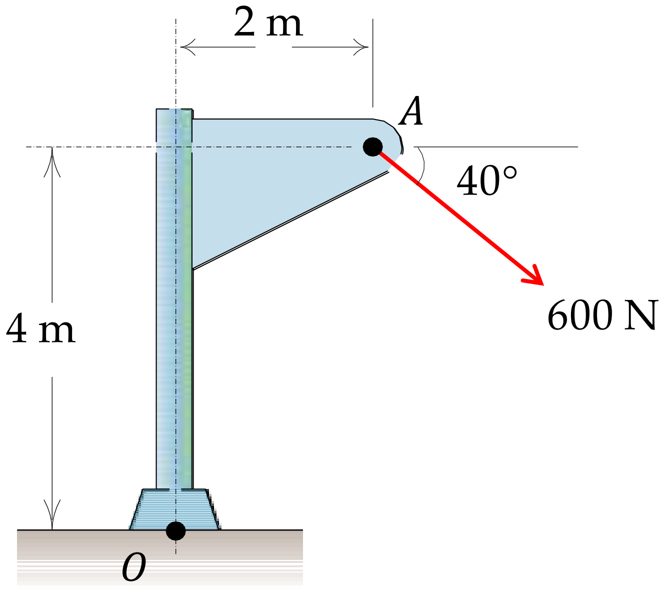
Using the distributive law for cross products

$$\boxed{\mathbf{M}_O = \mathbf{r} \times \mathbf{P} + \mathbf{r} \times \mathbf{Q}}$$

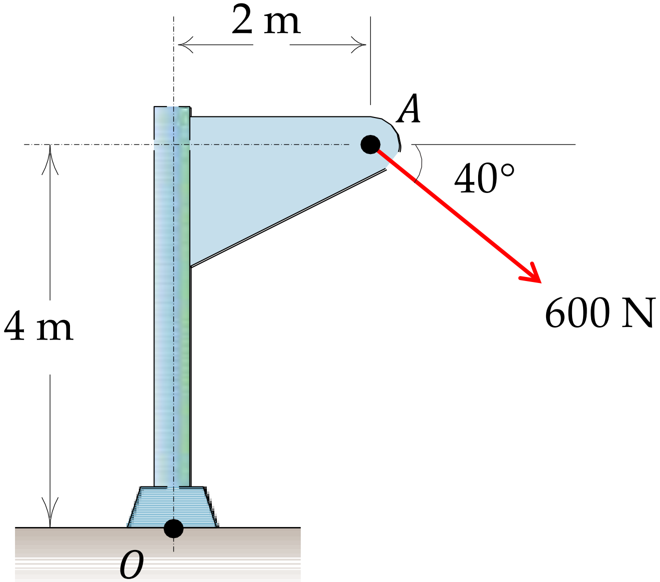
$$\boxed{M_O = Pp - Qq = -Rd}$$

# Example 1

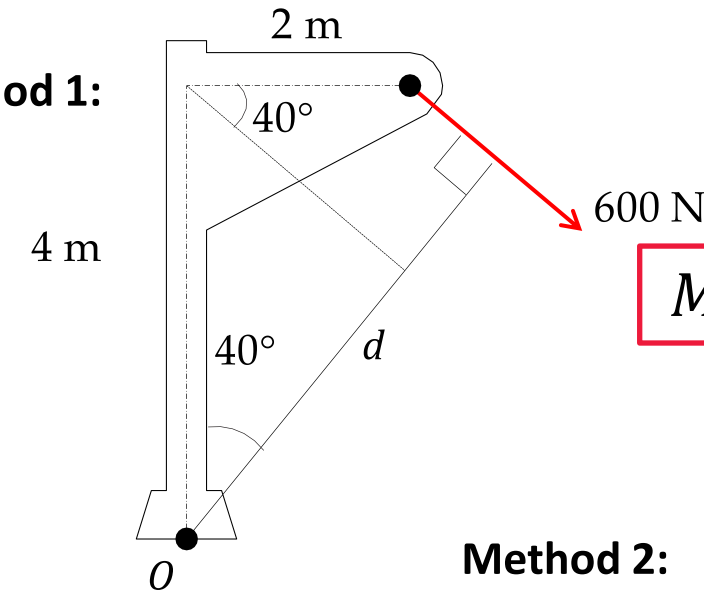
- Calculate the magnitude of the moment about the base point *O* of the 600-N force in five different ways.



# Example 1

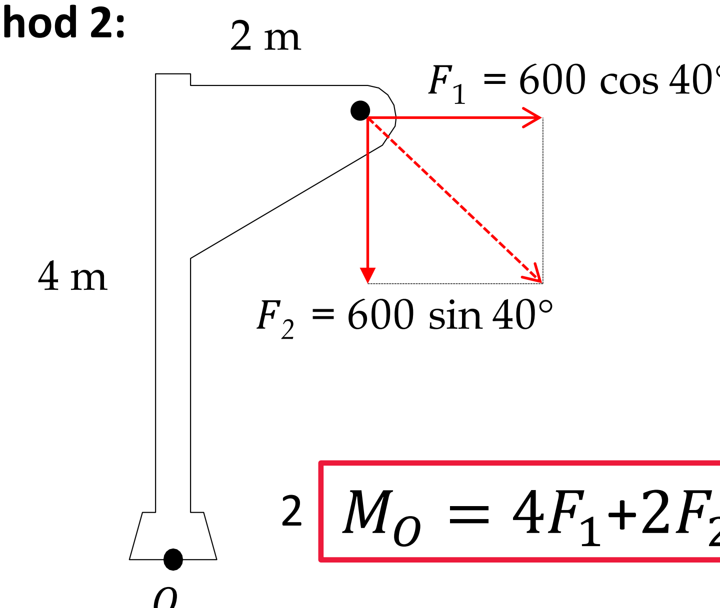



**Method 1:**



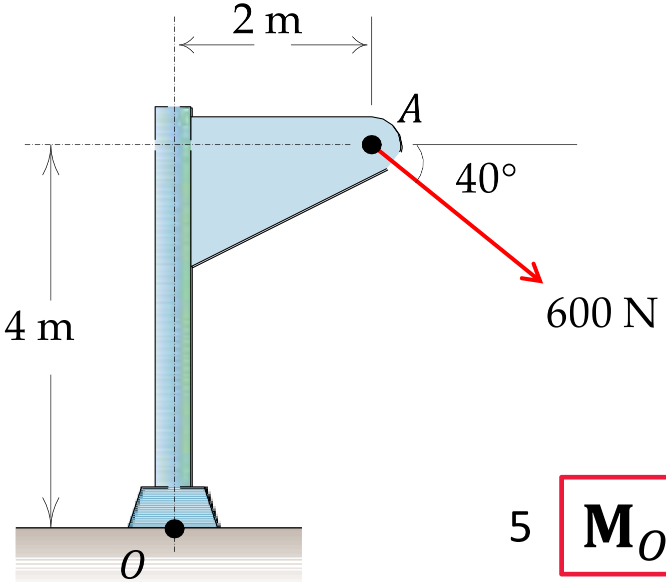

$$\boxed{M_O = Fd}$$

**Method 2:**




2 $$\boxed{M_O = 4F_1 + 2F_2}$$

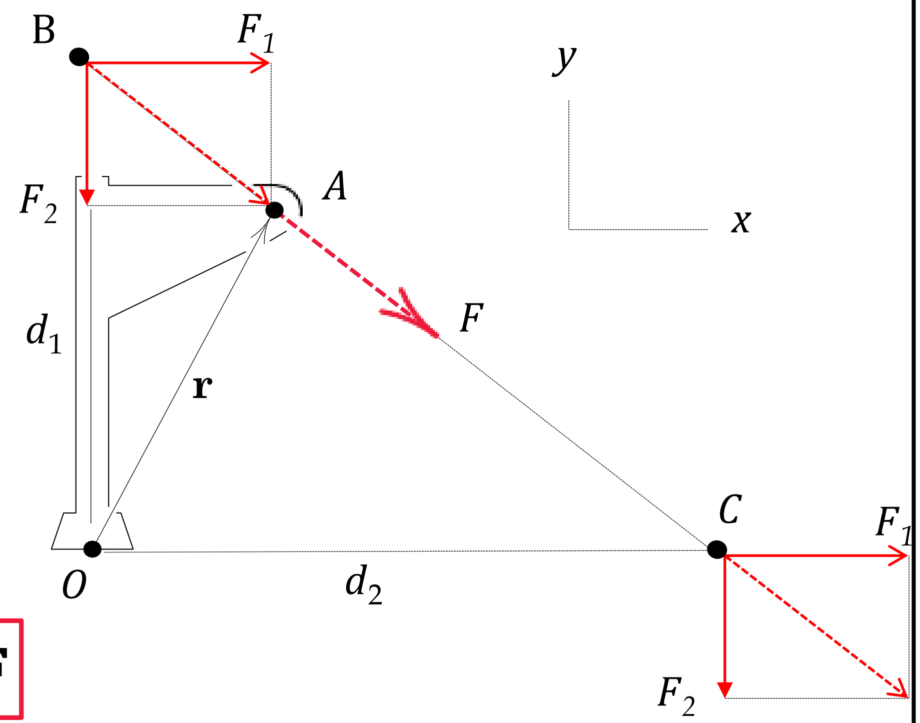
# Example 1




3 $\boxed{M_O = F_1 d_1}$

5 $\boxed{\mathbf{M}_O = \mathbf{r} \times \mathbf{F}}$

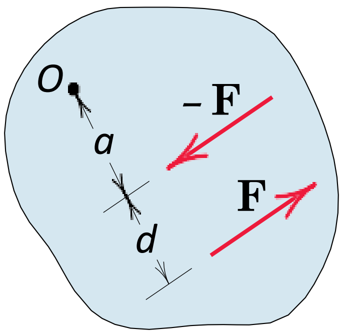
## Methods 3, 4 and 5:




4 $\boxed{M_O = F_2 d_2}$

# Couple

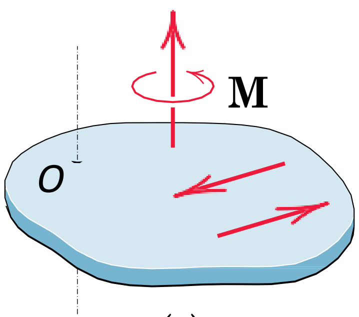
- The moment produced by two equal, opposite, and noncollinear forces is called a couple



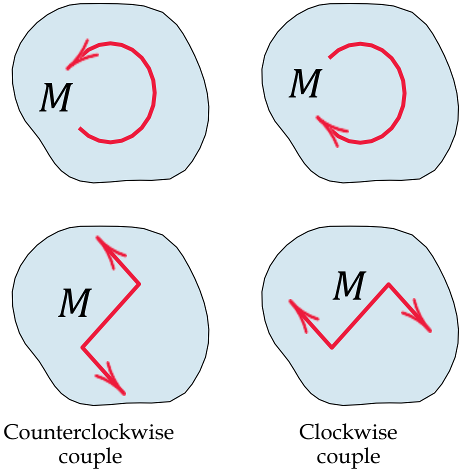
(a)

$$\boxed{M_O = F(a + d) - Fa}$$

$$\boxed{M_O = Fd}$$

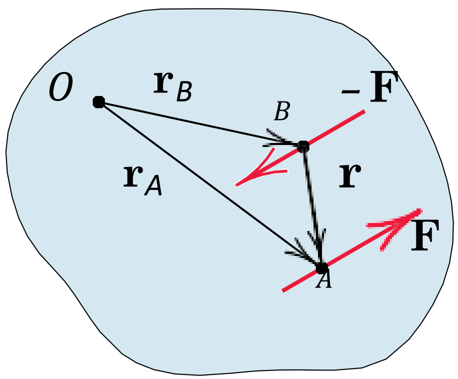


(c)



(d)

# Couple: A general case



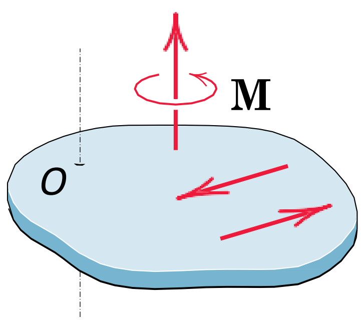

(b)

$$\boxed{\mathbf{M}_O = \mathbf{r}_A \times \mathbf{F} + \mathbf{r}_B \times -\mathbf{F}}$$

$$\boxed{\mathbf{M}_O = (\mathbf{r}_A - \mathbf{r}_B) \times \mathbf{F}}$$

$$\boxed{(\mathbf{r}_A - \mathbf{r}_B) = \mathbf{r}}$$

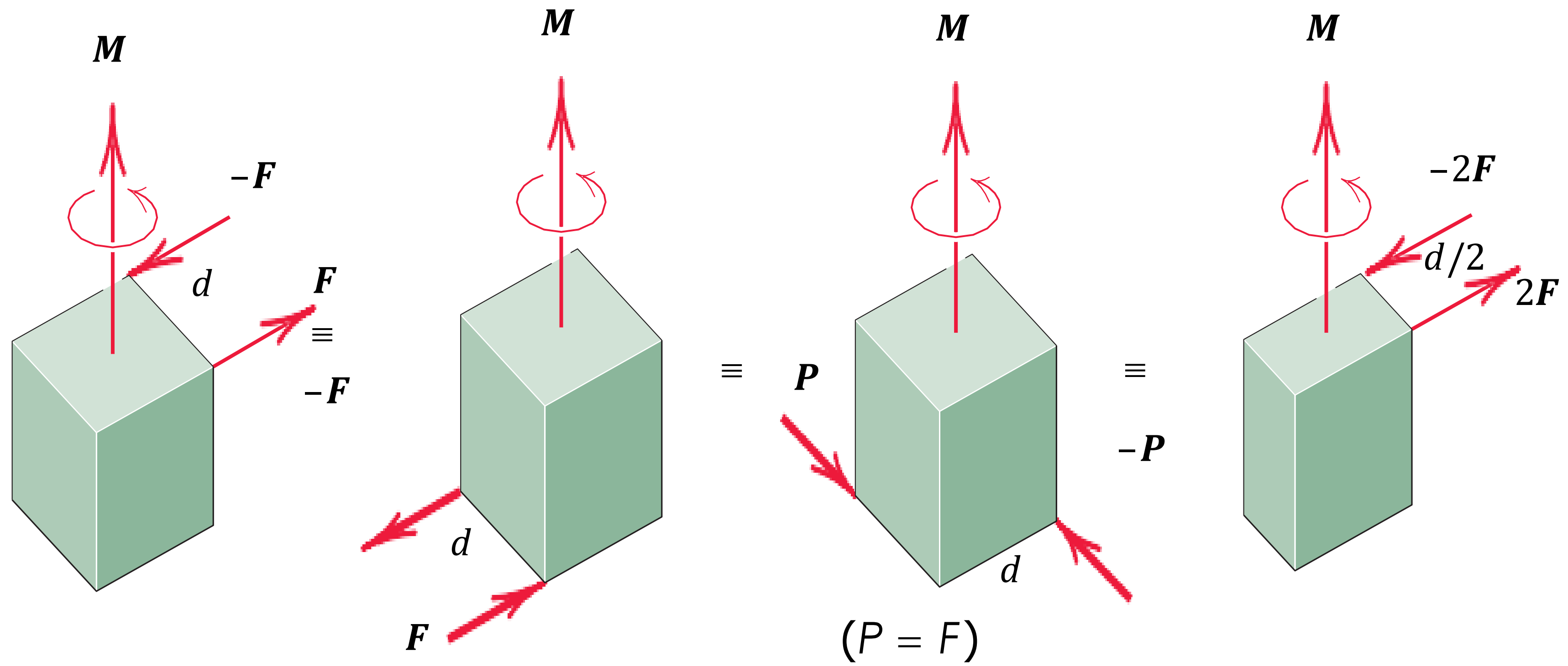
$$\boxed{\mathbf{M}_O = \mathbf{r} \times \mathbf{F}}$$



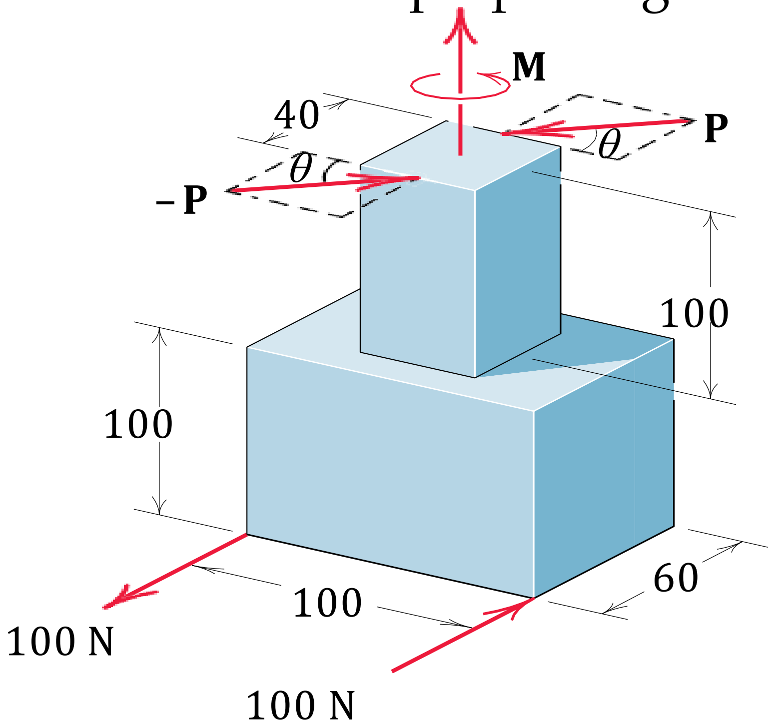

Independent from *O* and hence free vector

(c)

# Equivalent Couples

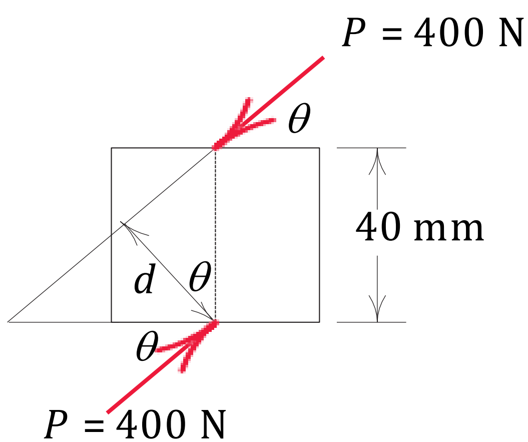


Example 2: The rigid structural member is subjected to a couple consisting of the two 100-N forces. Replace this couple by an equivalent couple consisting of the two forces P and -P, each of which has a magnitude of 400 N. Determine the proper angle .




Dimensions in millimeters

$$\boxed{M_O = 100 \times 0.1}$$

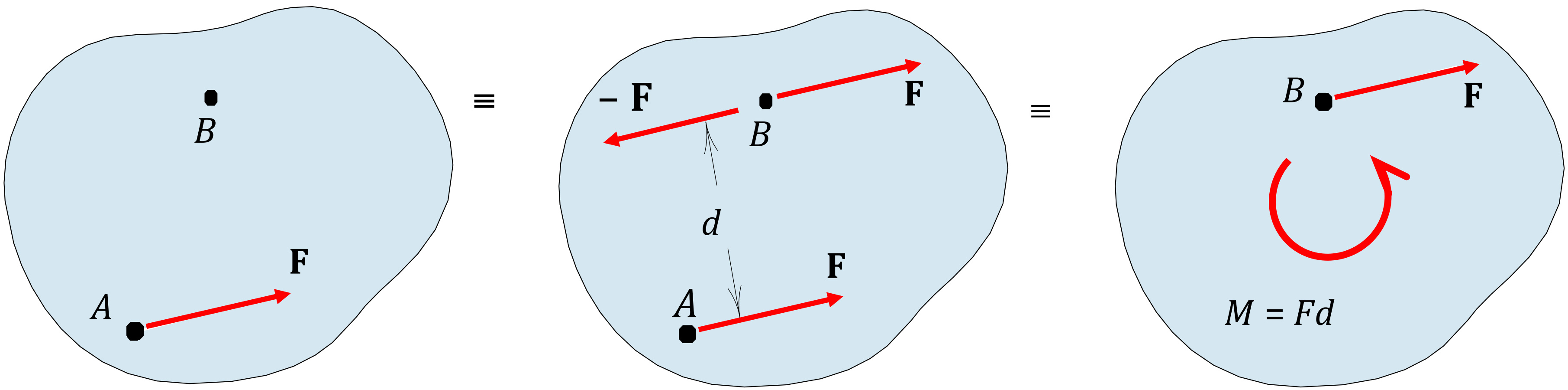



$$\boxed{M_O = 400 \times 0.04 \cos\theta}$$

Get $\theta$ by equating both equations

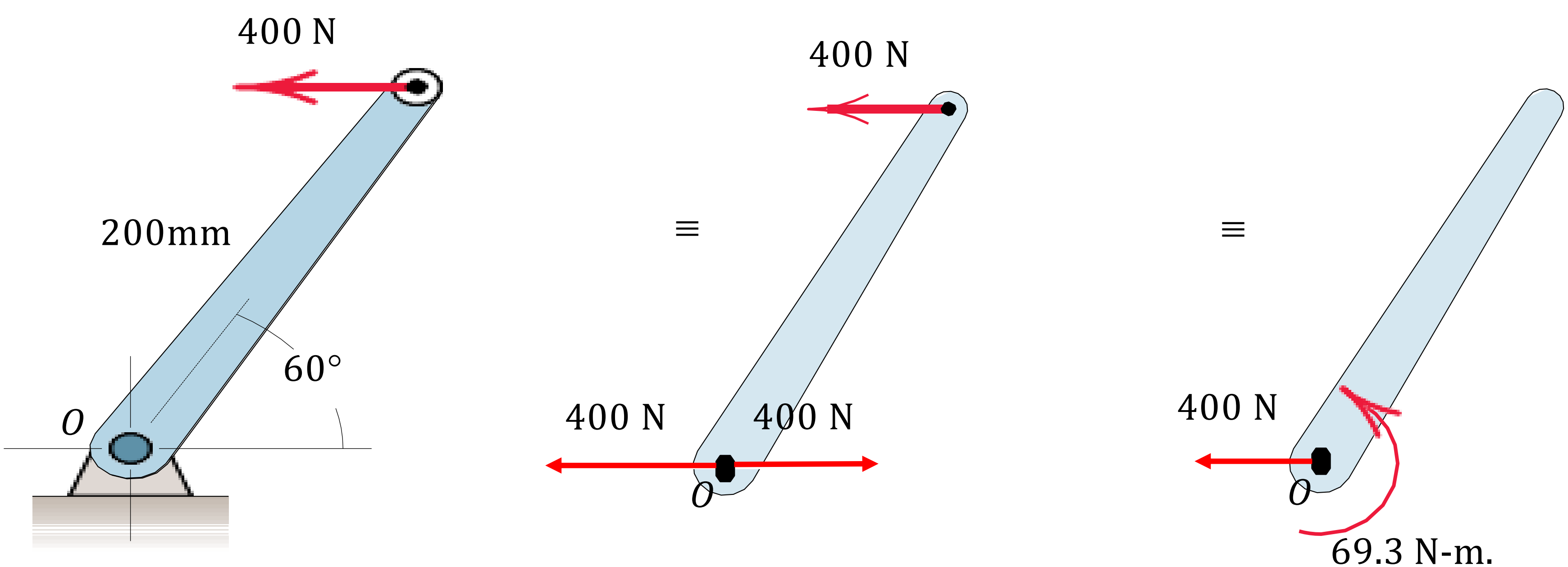
## Force–Couple Systems

- A force can be replaced of by a force and a couple and referred to as a force–couple system.




$$\boxed{M = Fd}$$

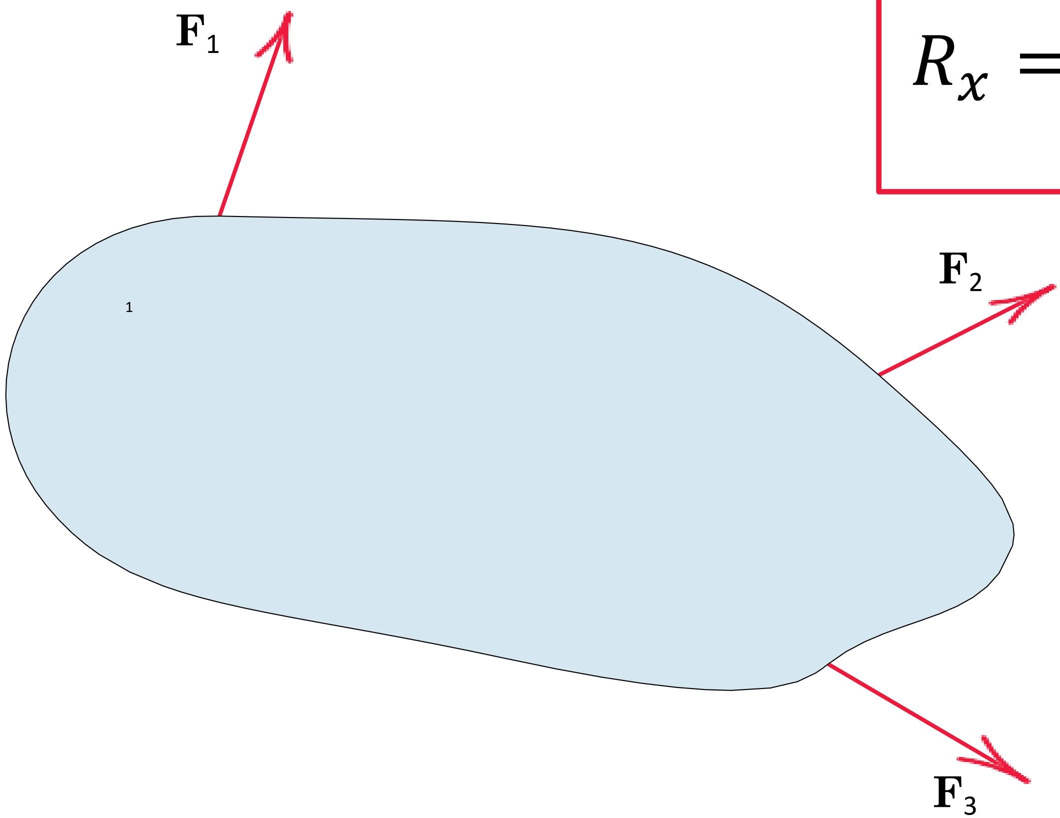
**Example 3:** Replace the horizontal 400-N force acting on the lever by an equivalent system consisting of a force at O and a couple.




$$\boxed{M_O = 400 \times 0.2 \sin 60}$$

# Resultant: Method 1

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \cdots + \mathbf{F}_n$$



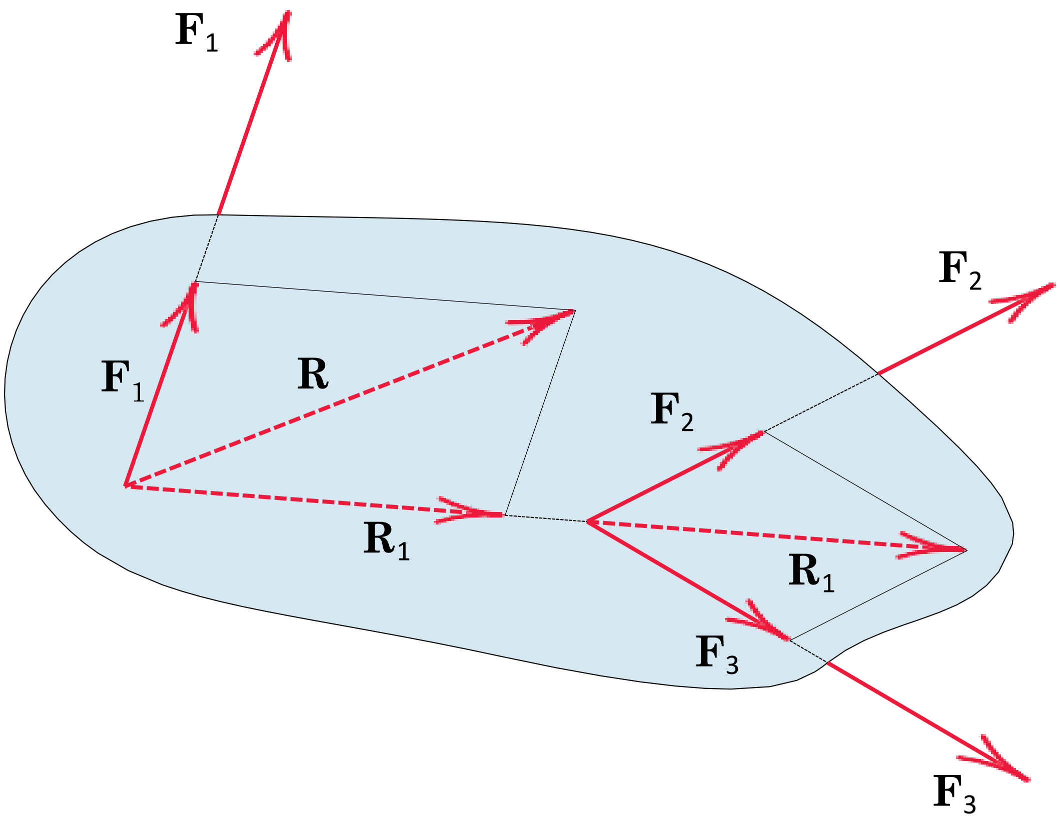

$$R_x = \sum F_{ix}$$

$$R_y = \sum F_{iy}$$

$$R = \sqrt{(R_x)^2 + \left(R_y\right)^2}$$

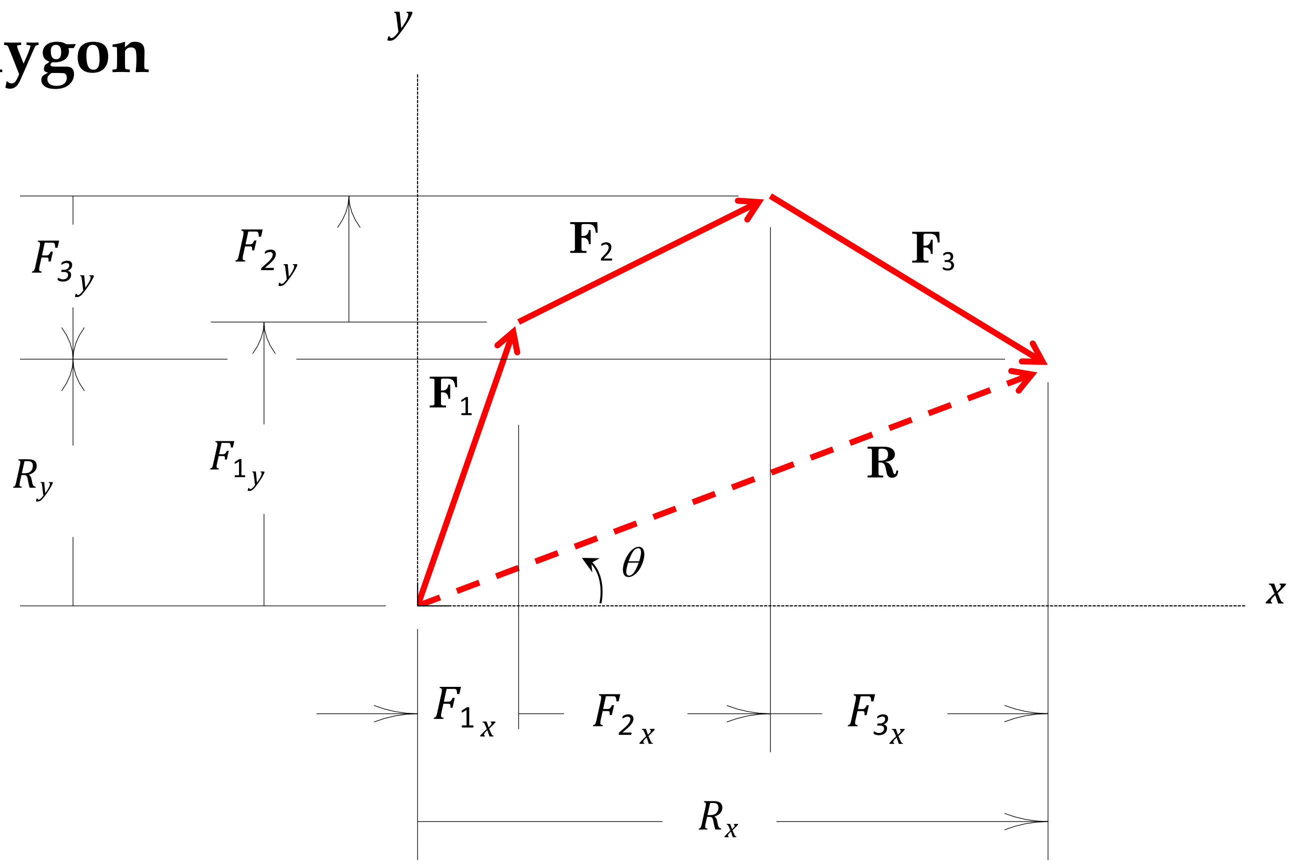
$$\theta = \tan^{-1}\left(R_y/R_x\right)$$

# Resultant: Method 2
# Parallelogram



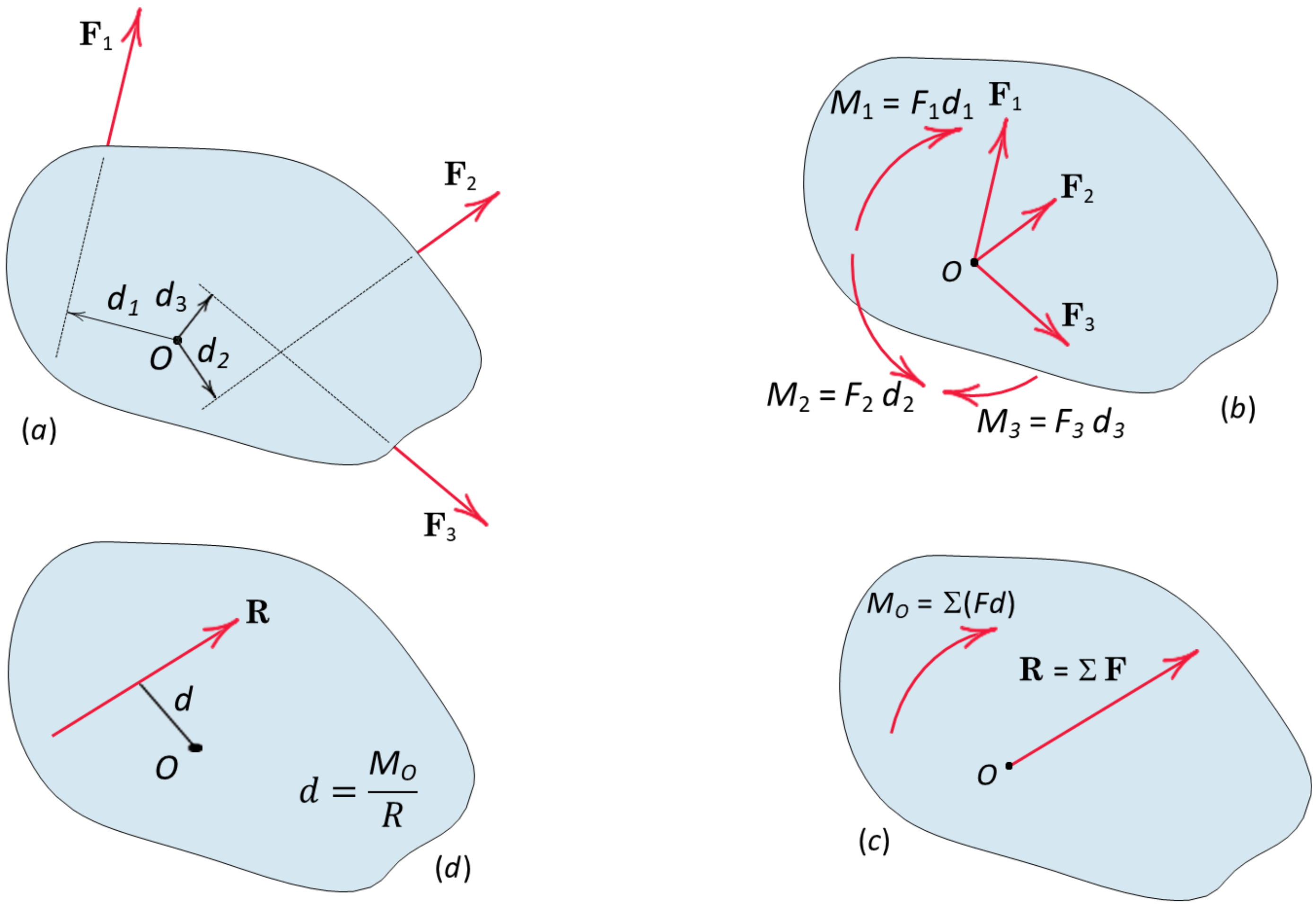

$(a)$

# Resultant: Method 2
# Force Polygon




(b)

# Resultant: Method 3



## Principle of Moments

- The moment of the resultant force about any point $O$ equals the sum of the moments of the original forces of the system about the same point.

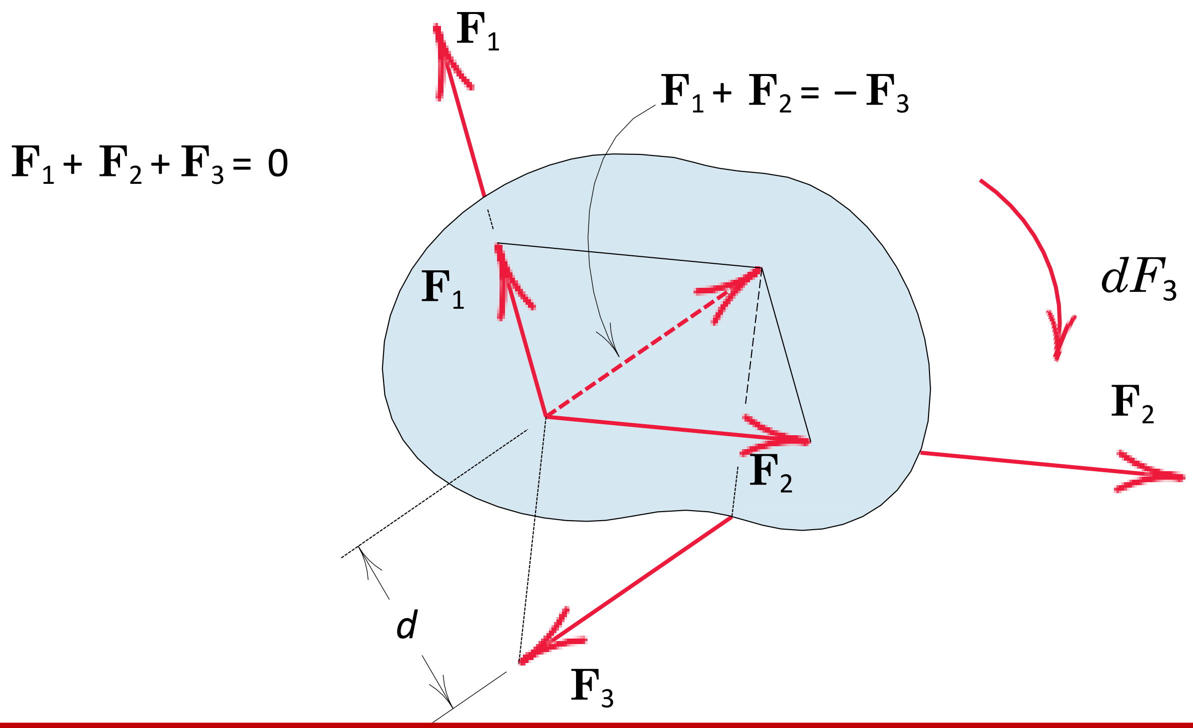
$$\boxed{Rd = M_O}$$

$$\boxed{M_O = \sum F_i d_i}$$

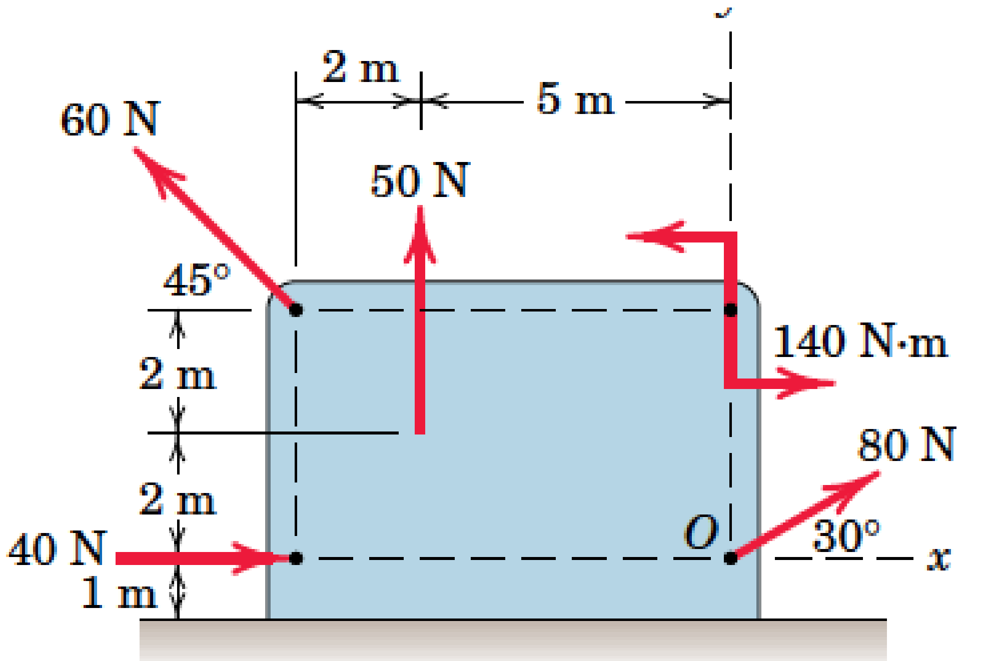
$$\boxed{\mathbf{R} = \sum \mathbf{F}_i}$$

# Resultant: An observation

- If the resultant force R for a given force system is zero, the resultant of the system need not be zero because the resultant may be a couple.

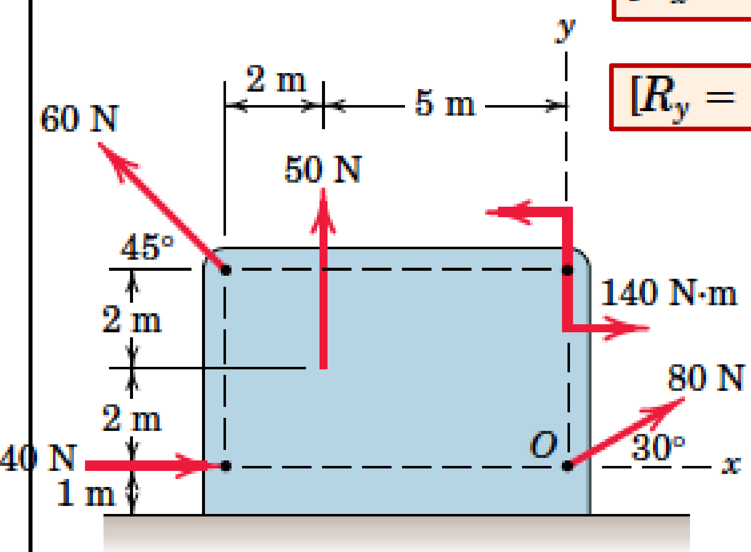


**Example 4:** Determine the resultant of the four forces and one couple which act on the plate shown.



# Example 4

## Part 1: Resultant

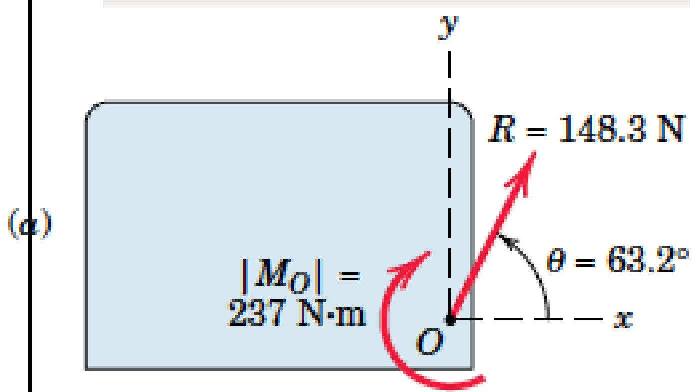



$[R_x = \Sigma F_x]$ $R_x = 40 + 80\cos 30^\circ - 60\cos 45^\circ = 66.9\text{ N}$

$[R_y = \Sigma F_y]$ $R_y = 50 + 80\sin 30^\circ + 60\cos 45^\circ = 132.4\text{ N}$

$$R = \sqrt{(66.9)^2 + (132.4)^2} = 148.3\text{ N}$$

$$\theta = \tan^{-1}\frac{132.4}{66.9} = 63.2^\circ$$

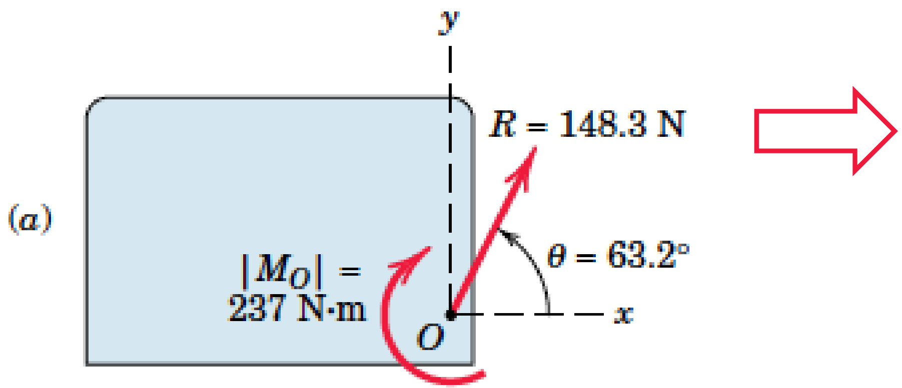


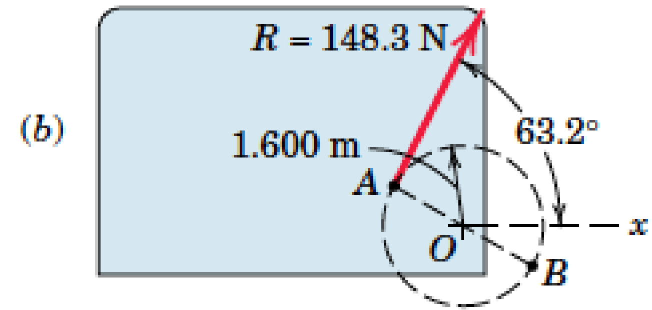

Part 2: Moment

$[M_O = \Sigma(Fd)]$

$$M_O = 140 - 50(5) + 60\cos 45^\circ(4) - 60\sin 45^\circ(7)$$
$$= -237\text{ N}\cdot\text{m}$$

# Example 4



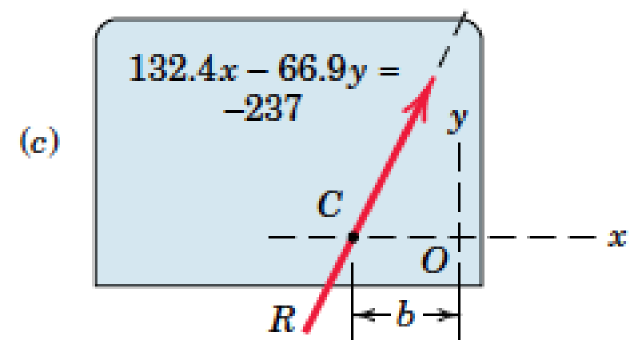




Part 3: Replacing force and moment by a force

$[Rd = |M_O|]$

$148.3d = 237$

$d = 1.600 \text{ m}$




Part 4: Intercept of force on x axis

$R_y b = |M_O|$

$b = \dfrac{237}{132.4} = 1.792 \text{ m}$

# THANK YOU